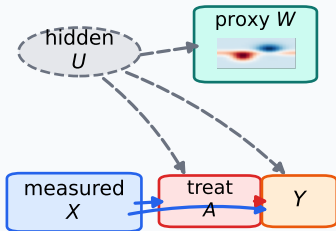


Observed system



Observed data: (X, W, A, Y)

Target: effect of A on Y

Image representation



Key condition:
 R_W preserves the U signal

Proxy balancing

Balance on (R_X, R_W)

treated



controls

weighted
controls
match

$\hat{\tau}$ for $A \rightarrow Y$

Then latent imbalance in U
shrinks without observing U

Single Proxy Balancing: use image-derived proxy representations to balance hidden severity, not just observed covariates.